분산시스템 과제 #8

박종현, 2025-05-22
공과대학 컴퓨터정보통신공학과

과제 목표

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[ch 3-13]

```
hduser@dist:~$ cd data
hduser@dist:~/data$ ls
hduser@dist:~/data$ vi emp.cs
hduser@dist:~/data$ vi emp.csv
hduser@dist:~/data$ dhfs dfs -put emp.csv
Command 'dhfs' not found, did you mean:
  command 'xhfs' from deb hfsutils-tcltk (3.2.6-15build2)
  command 'hdfs' from deb hdfs-cli (2.3.0-2ubuntu0.24.04.2)
  command 'hfs' from deb hfsutils-tcltk (3.2.6-15build2)
Try: sudo apt install <deb name>
hduser@dist:~/data$ hdfs dfs -put emp.csv /output
hduser@dist:~/data$ echo "create 'employees', 'personal', 'department'" | hbase shell
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/home/hduser/hbase-2.6.2-hadoop3/lib/client-facing-thirdparty/log4j-slf4j-impl-2.17.2.
jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/home/hduser/hadoop-3.4.1/share/hadoop/common/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j
/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
HBase Shell
Use "help" to get list of supported commands.
Use "exit" to quit this interactive shell.
For Reference, please visit: http://hbase.apache.org/2.0/book.html#shell
Version 2.6.2-hadoop3, r6b3b36b429cf9a9d74110de79eb3b327b29ebf17, Fri Feb 14 14:59:52 UTC 2025
Took 0.0152 seconds
hbase:001:0> create 'employees', 'personal', 'department'
Created table employees
Took 3.9686 seconds
=> Hbase::Table - employees
hbase:002:0>
hduser@dist:~/data$
```

```
                HDFS: Number of write operations=0
                HDFS: Number of bytes read erasure-coded=0
        Job Counters
                Launched map tasks=1
                Data-local map tasks=1
                Total time spent by all maps in occupied slots (ms)=11196
                Total time spent by all reduces in occupied slots (ms)=0
                Total time spent by all map tasks (ms)=11196
                Total vcore-milliseconds taken by all map tasks=11196
                Total megabyte-milliseconds taken by all map tasks=11464704
        Map-Reduce Framework
                Map input records=4
                Map output records=4
                Input split bytes=101
                Spilled Records=0
                Failed Shuffles=0
                Merged Map outputs=0
                GC time elapsed (ms)=402
                CPU time spent (ms)=7080
                Physical memory (bytes) snapshot=417566720
                Virtual memory (bytes) snapshot=2607124480
                Total committed heap usage (bytes)=313524224
                Peak Map Physical memory (bytes)=417566720
                Peak Map Virtual memory (bytes)=2607124480
        ImportTsv
                Bad Lines=0
        File Input Format Counters
                Bytes Read=76
        File Output Format Counters
                Bytes Written=0
hduser@dist:~/data$
```

```
                Physical memory (bytes) snapshot=417566720
                Virtual memory (bytes) snapshot=2607124480
                Total committed heap usage (bytes)=313524224
                Peak Map Physical memory (bytes)=417566720
                Peak Map Virtual memory (bytes)=2607124480
        ImportTsv
                Bad Lines=0
        File Input Format Counters
                Bytes Read=76
        File Output Format Counters
                Bytes Written=0
hduser@dist:~/data$ echo "create 'employees_copy', 'personal', 'department'" | hbase shell
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/home/hduser/hbase-2.6.2-hadoop3/lib/client-facing-thirdparty/log4j-slf4j-impl-2.17.2.
jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/home/hduser/hadoop-3.4.1/share/hadoop/common/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j
/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
HBase Shell
Use "help" to get list of supported commands.
Use "exit" to quit this interactive shell.
For Reference, please visit: http://hbase.apache.org/2.0/book.html#shell
Version 2.6.2-hadoop3, r6b3b36b429cf9a9d74110de79eb3b327b29ebf17, Fri Feb 14 14:59:52 UTC 2025
Took 0.0121 seconds
hbase:001:0> create 'employees_copy', 'personal', 'department'
Created table employees_copy
Took 3.8757 seconds
=> Hbase::Table - employees_copy
hbase:002:0>
hduser@dist:~/data$
```

```
                Map output records=4
                Input split bytes=101
                Spilled Records=0
                Failed Shuffles=0
                Merged Map outputs=0
                GC time elapsed (ms)=269
                CPU time spent (ms)=7300
                Physical memory (bytes) snapshot=443363328
                Virtual memory (bytes) snapshot=2588397568
                Total committed heap usage (bytes)=314572800
                Peak Map Physical memory (bytes)=443510784
                Peak Map Virtual memory (bytes)=2588397568
        HBaseCounters
                BLOCK_BYTES_SCANNED=0
                BYTES_IN_REMOTE_RESULTS=0
                BYTES_IN_RESULTS=364
                FS_READ_TIME=0
                MILLIS_BETWEEN_NEXTS=2373
                NOT_SERVING_REGION_EXCEPTION=0
                REGIONS_SCANNED=1
                REMOTE_RPC_CALLS=0
                REMOTE_RPC_RETRIES=0
                ROWS_FILTERED=0
                ROWS_SCANNED=4
                RPC_CALLS=1
                RPC_RETRIES=0
        File Input Format Counters
                Bytes Read=0
        File Output Format Counters
                Bytes Written=0
hduser@dist:~/data$
```

```
                Map output records=4
                Input split bytes=101
                Spilled Records=0
                Failed Shuffles=0
                Merged Map outputs=0
                GC time elapsed (ms)=208
                CPU time spent (ms)=6140
                Physical memory (bytes) snapshot=438149120
                Virtual memory (bytes) snapshot=2575482880
                Total committed heap usage (bytes)=316669952
                Peak Map Physical memory (bytes)=438149120
                Peak Map Virtual memory (bytes)=2575482880
        HBaseCounters
                BLOCK_BYTES_SCANNED=0
                BYTES_IN_REMOTE_RESULTS=0
                BYTES_IN_RESULTS=364
                FS_READ_TIME=0
                MILLIS_BETWEEN_NEXTS=1681
                NOT_SERVING_REGION_EXCEPTION=0
                REGIONS_SCANNED=1
                REMOTE_RPC_CALLS=0
                REMOTE_RPC_RETRIES=0
                ROWS_FILTERED=0
                ROWS_SCANNED=4
                RPC_CALLS=1
                RPC_RETRIES=0
        File Input Format Counters
                Bytes Read=0
        File Output Format Counters
                Bytes Written=526
hduser@dist:~/data$
```

```
                REGIONS_SCANNED=1
                REMOTE_RPC_CALLS=0
                REMOTE_RPC_RETRIES=0
                ROWS_FILTERED=0
                ROWS_SCANNED=4
                RPC_CALLS=1
                RPC_RETRIES=0
        File Input Format Counters
                Bytes Read=0
        File Output Format Counters
                Bytes Written=526
hduser@dist:~/data$ echo "create 'employees_import', 'personal', 'department'" | hbase shell
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/home/hduser/hbase-2.6.2-hadoop3/lib/client-facing-thirdparty/log4j-slf4j-impl-2.17.2.
jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/home/hduser/hadoop-3.4.1/share/hadoop/common/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j
/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
HBase Shell
Use "help" to get list of supported commands.
Use "exit" to quit this interactive shell.
For Reference, please visit: http://hbase.apache.org/2.0/book.html#shell
Version 2.6.2-hadoop3, r6b3b36b429cf9a9d74110de79eb3b327b29ebf17, Fri Feb 14 14:59:52 UTC 2025
Took 0.0043 seconds
hbase:001:0> create 'employees_import', 'personal', 'department'
Created table employees_import
Took 4.1485 seconds
=> Hbase::Table - employees_import
hbase:002:0>
hduser@dist:~/data$
```

```
2025-05-21T18:04:50,604 INFO  [main {}] mapreduce.Job:
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=0
                HDFS: Number of bytes read erasure-coded=0
        Job Counters
                Launched map tasks=1
                Data-local map tasks=1
                Total time spent by all maps in occupied slots (ms)=10145
                Total time spent by all reduces in occupied slots (ms)=0
                Total time spent by all map tasks (ms)=10145
                Total vcore-milliseconds taken by all map tasks=10145
                Total megabyte-milliseconds taken by all map tasks=10388480
        Map-Reduce Framework
                Map input records=4
                Map output records=4
                Input split bytes=116
                Spilled Records=0
                Failed Shuffles=0
                Merged Map outputs=0
                GC time elapsed (ms)=296
                CPU time spent (ms)=7790
                Physical memory (bytes) snapshot=352215040
                Virtual memory (bytes) snapshot=2595033088
                Total committed heap usage (bytes)=240123904
                Peak Map Physical memory (bytes)=352215040
                Peak Map Virtual memory (bytes)=2595033088
        File Input Format Counters
                Bytes Read=526
        File Output Format Counters
                Bytes Written=0
hduser@dist:~/data$
```

[ch 3-14]

```
                Spilled Records=0
                Failed Shuffles=0
                Merged Map outputs=0
                GC time elapsed (ms)=201
                CPU time spent (ms)=6250
                Physical memory (bytes) snapshot=427003904
                Virtual memory (bytes) snapshot=2589220864
                Total committed heap usage (bytes)=316669952
                Peak Map Physical memory (bytes)=427003904
                Peak Map Virtual memory (bytes)=2589220864
        HBaseCounters
                BLOCK_BYTES_SCANNED=0
                BYTES_IN_REMOTE_RESULTS=0
                BYTES_IN_RESULTS=195
                FS_READ_TIME=0
                MILLIS_BETWEEN_NEXTS=1951
                NOT_SERVING_REGION_EXCEPTION=0
                REGIONS_SCANNED=1
                REMOTE_RPC_CALLS=0
                REMOTE_RPC_RETRIES=0
                ROWS_FILTERED=0
                ROWS_SCANNED=4
                RPC_CALLS=1
                RPC_RETRIES=0
        org.apache.hadoop.hbase.mapreduce.RowCounter$RowCounterMapper$Counters
                ROWS=4
        File Input Format Counters
                Bytes Read=0
        File Output Format Counters
                Bytes Written=0
hduser@dist:~/data$
```

```
                BYTES_IN_REMOTE_RESULTS=0
                BYTES_IN_RESULTS=364
                FS_READ_TIME=0
                MILLIS_BETWEEN_NEXTS=3848
                NOT_SERVING_REGION_EXCEPTION=0
                REGIONS_SCANNED=1
                REMOTE_RPC_CALLS=0
                REMOTE_RPC_RETRIES=0
                ROWS_FILTERED=0
                ROWS_SCANNED=4
                RPC_CALLS=1
                RPC_RETRIES=0
        Q
                department:role_Size=179
                personal:name_Size=153
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        org.apache.hadoop.hbase.mapreduce.CellCounter$CellCounterMapper$Counters
                CELLS=8
                ROWS=4
                SIZE=332
        File Input Format Counters
                Bytes Read=0
        File Output Format Counters
                Bytes Written=464
hduser@dist:~/data$
```

```
hduser@dist:~/data$ hdfs dfs -cat /output/emp_cells/part-r-00000
1:department:role_Versions      1
1:personal:name_Versions        1
2:department:role_Versions      1
2:personal:name_Versions        1
3:department:role_Versions      1
3:personal:name_Versions        1
4:department:role_Versions      1
4:personal:name_Versions        1
Total Families Across all Rows  2
Total Qualifiers across all Rows        8
Total ROWS      4
Total SIZE      332
department      1
department:role 4
department:role_Size    179
department_Size 179
personal        1
personal:name   4
personal:name_Size      153
personal_Size   153
hduser@dist:~/data$
```

```
98)) - Client environment:user.dir=/home/hduser/data
2025-05-21 18:09:05,872 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ZooKeeper (Environment.java:logEnv(
98)) - Client environment:os.memory.free=75MB
2025-05-21 18:09:05,874 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ZooKeeper (Environment.java:logEnv(
98)) - Client environment:os.memory.max=871MB
2025-05-21 18:09:05,874 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ZooKeeper (Environment.java:logEnv(
98)) - Client environment:os.memory.total=106MB
2025-05-21 18:09:05,891 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ZooKeeper (ZooKeeper.java:<init>(63
7)) - Initiating client connection, connectString=127.0.0.1:2181 sessionTimeout=90000 watcher=org.apache.hadoop.hbase.zo
okeeper.ReadOnlyZKClient$$Lambda$22/1030257@519758a3
2025-05-21 18:09:05,910 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ClientCnxnSocket (ClientCnxnSocket.
java:initProperties(239)) - jute.maxbuffer value is 1048575 Bytes
2025-05-21 18:09:05,939 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ClientCnxn (ClientCnxn.java:initReq
uestTimeout(1747)) - zookeeper.request.timeout value is 0. feature enabled=false
2025-05-21 18:09:05,977 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:logStartConnect(1177)) - Opening socket connection to server localhost/127.0.0.1:2181.
2025-05-21 18:09:05,977 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:logStartConnect(1179)) - SASL config status: Will not attempt to authenticate using SASL (unknown er
ror)
2025-05-21 18:09:05,994 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:primeConnection(1013)) - Socket connection established, initiating session, client: /127.0.0.1:57010
, server: localhost/127.0.0.1:2181
2025-05-21 18:09:06,012 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:onConnected(1453)) - Session establishment complete on server localhost/127.0.0.1:2181, session id =
 0x10000292f1b0019, negotiated timeout = 90000
2025-05-21 18:09:10,764 INFO  [main] client.HBaseAdmin (HBaseAdmin.java:postOperationResult(3751)) - Operation: CREATE,
Table Name: default:testtable, procId: 24 completed
2025-05-21 18:09:10,767 INFO  [main] client.ConnectionImplementation (ConnectionImplementation.java:closeMasterService(2
127)) - Closing master protocol: MasterService
testtable is created
hduser@dist:~/data$
```

```
2025-05-21 18:09:32,757 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ZooKeeper (Environment.java:logEnv(
98)) - Client environment:user.name=hduser
2025-05-21 18:09:32,757 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ZooKeeper (Environment.java:logEnv(
98)) - Client environment:user.home=/home/hduser
2025-05-21 18:09:32,757 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ZooKeeper (Environment.java:logEnv(
98)) - Client environment:user.dir=/home/hduser/data
2025-05-21 18:09:32,757 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ZooKeeper (Environment.java:logEnv(
98)) - Client environment:os.memory.free=75MB
2025-05-21 18:09:32,758 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ZooKeeper (Environment.java:logEnv(
98)) - Client environment:os.memory.max=871MB
2025-05-21 18:09:32,758 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ZooKeeper (Environment.java:logEnv(
98)) - Client environment:os.memory.total=106MB
2025-05-21 18:09:32,783 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ZooKeeper (ZooKeeper.java:<init>(63
7)) - Initiating client connection, connectString=127.0.0.1:2181 sessionTimeout=90000 watcher=org.apache.hadoop.hbase.zo
okeeper.ReadOnlyZKClient$$Lambda$22/288093241@2a3a48a5
2025-05-21 18:09:32,793 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ClientCnxnSocket (ClientCnxnSocket.
java:initProperties(239)) - jute.maxbuffer value is 1048575 Bytes
2025-05-21 18:09:32,825 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ClientCnxn (ClientCnxn.java:initReq
uestTimeout(1747)) - zookeeper.request.timeout value is 0. feature enabled=false
2025-05-21 18:09:32,857 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:logStartConnect(1177)) - Opening socket connection to server localhost/127.0.0.1:2181.
2025-05-21 18:09:32,858 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:logStartConnect(1179)) - SASL config status: Will not attempt to authenticate using SASL (unknown er
ror)
2025-05-21 18:09:32,876 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:primeConnection(1013)) - Socket connection established, initiating session, client: /127.0.0.1:54852
, server: localhost/127.0.0.1:2181
2025-05-21 18:09:32,906 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:onConnected(1453)) - Session establishment complete on server localhost/127.0.0.1:2181, session id =
 0x10000292f1b001a, negotiated timeout = 90000
Put is completed
```

```
2025-05-21 18:09:43,281 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ZooKeeper (Environment.java:logEnv(
98)) - Client environment:os.memory.free=75MB
2025-05-21 18:09:43,282 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ZooKeeper (Environment.java:logEnv(
98)) - Client environment:os.memory.max=871MB
2025-05-21 18:09:43,288 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ZooKeeper (Environment.java:logEnv(
98)) - Client environment:os.memory.total=106MB
2025-05-21 18:09:43,296 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ZooKeeper (ZooKeeper.java:<init>(63
7)) - Initiating client connection, connectString=127.0.0.1:2181 sessionTimeout=90000 watcher=org.apache.hadoop.hbase.zo
okeeper.ReadOnlyZKClient$$Lambda$22/1121142575@1d857dc5
2025-05-21 18:09:43,309 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ClientCnxnSocket (ClientCnxnSocket.
java:initProperties(239)) - jute.maxbuffer value is 1048575 Bytes
2025-05-21 18:09:43,326 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ClientCnxn (ClientCnxn.java:initReq
uestTimeout(1747)) - zookeeper.request.timeout value is 0. feature enabled=false
2025-05-21 18:09:43,367 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:logStartConnect(1177)) - Opening socket connection to server localhost/127.0.0.1:2181.
2025-05-21 18:09:43,368 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:logStartConnect(1179)) - SASL config status: Will not attempt to authenticate using SASL (unknown er
ror)
2025-05-21 18:09:43,376 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:primeConnection(1013)) - Socket connection established, initiating session, client: /127.0.0.1:32946
, server: localhost/127.0.0.1:2181
2025-05-21 18:09:43,395 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:onConnected(1453)) - Session establishment complete on server localhost/127.0.0.1:2181, session id =
 0x10000292f1b001b, negotiated timeout = 90000
Found row: keyvalues={key1/cf1:a/1747850974695/Put/vlen=4/seqid=0, key1/cf1:b/1747850974695/Put/vlen=4/seqid=0}
Found row: keyvalues={key2/cf1:a/1747850974714/Put/vlen=4/seqid=0, key2/cf1:c/1747850974714/Put/vlen=4/seqid=0}
Scan is completed
cf1:a - Key : key1, Retrived value : val1
cf1:b - Key : key1, Retrived value : val2
keyvalues={key2/cf1:a/1747850974714/Put/vlen=4/seqid=0, key2/cf1:c/1747850974714/Put/vlen=4/seqid=0}
Get is completed
```

```
98)) - Client environment:user.name=hduser
2025-05-21 18:09:51,945 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ZooKeeper (Environment.java:logEnv(
98)) - Client environment:user.home=/home/hduser
2025-05-21 18:09:51,945 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ZooKeeper (Environment.java:logEnv(
98)) - Client environment:user.dir=/home/hduser/data
2025-05-21 18:09:51,945 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ZooKeeper (Environment.java:logEnv(
98)) - Client environment:os.memory.free=75MB
2025-05-21 18:09:51,945 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ZooKeeper (Environment.java:logEnv(
98)) - Client environment:os.memory.max=871MB
2025-05-21 18:09:51,945 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ZooKeeper (Environment.java:logEnv(
98)) - Client environment:os.memory.total=106MB
2025-05-21 18:09:51,956 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ZooKeeper (ZooKeeper.java:<init>(63
7)) - Initiating client connection, connectString=127.0.0.1:2181 sessionTimeout=90000 watcher=org.apache.hadoop.hbase.zo
okeeper.ReadOnlyZKClient$$Lambda$22/970818114@11df9173
2025-05-21 18:09:51,962 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ClientCnxnSocket (ClientCnxnSocket.
java:initProperties(239)) - jute.maxbuffer value is 1048575 Bytes
2025-05-21 18:09:51,988 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ClientCnxn (ClientCnxn.java:initReq
uestTimeout(1747)) - zookeeper.request.timeout value is 0. feature enabled=false
2025-05-21 18:09:52,009 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:logStartConnect(1177)) - Opening socket connection to server localhost/127.0.0.1:2181.
2025-05-21 18:09:52,010 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:logStartConnect(1179)) - SASL config status: Will not attempt to authenticate using SASL (unknown er
ror)
2025-05-21 18:09:52,023 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:primeConnection(1013)) - Socket connection established, initiating session, client: /127.0.0.1:51660
, server: localhost/127.0.0.1:2181
2025-05-21 18:09:52,055 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:onConnected(1453)) - Session establishment complete on server localhost/127.0.0.1:2181, session id =
 0x10000292f1b001c, negotiated timeout = 90000
List Update is completed
hduser@dist:~/data$
```

```
2025-05-21 18:11:09,455 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ZooKeeper (Environment.java:logEnv(
98)) - Client environment:user.home=/home/hduser
2025-05-21 18:11:09,455 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ZooKeeper (Environment.java:logEnv(
98)) - Client environment:user.dir=/home/hduser/data
2025-05-21 18:11:09,456 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ZooKeeper (Environment.java:logEnv(
98)) - Client environment:os.memory.free=75MB
2025-05-21 18:11:09,456 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ZooKeeper (Environment.java:logEnv(
98)) - Client environment:os.memory.max=871MB
2025-05-21 18:11:09,456 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ZooKeeper (Environment.java:logEnv(
98)) - Client environment:os.memory.total=106MB
2025-05-21 18:11:09,463 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ZooKeeper (ZooKeeper.java:<init>(63
7)) - Initiating client connection, connectString=127.0.0.1:2181 sessionTimeout=90000 watcher=org.apache.hadoop.hbase.zo
okeeper.ReadOnlyZKClient$$Lambda$22/1293817201@7ea64634
2025-05-21 18:11:09,472 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ClientCnxnSocket (ClientCnxnSocket.
java:initProperties(239)) - jute.maxbuffer value is 1048575 Bytes
2025-05-21 18:11:09,488 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ClientCnxn (ClientCnxn.java:initReq
uestTimeout(1747)) - zookeeper.request.timeout value is 0. feature enabled=false
2025-05-21 18:11:09,513 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:logStartConnect(1177)) - Opening socket connection to server localhost/127.0.0.1:2181.
2025-05-21 18:11:09,513 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:logStartConnect(1179)) - SASL config status: Will not attempt to authenticate using SASL (unknown er
ror)
2025-05-21 18:11:09,529 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:primeConnection(1013)) - Socket connection established, initiating session, client: /127.0.0.1:49038
, server: localhost/127.0.0.1:2181
2025-05-21 18:11:09,555 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:onConnected(1453)) - Session establishment complete on server localhost/127.0.0.1:2181, session id =
 0x10000292f1b001d, negotiated timeout = 90000
Number of rows : 2
Delete is completed
hduser@dist:~/data$
```

```
2025-05-21 18:11:29,588 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ZooKeeper (Environment.java:logEnv(
98)) - Client environment:os.memory.free=74MB
2025-05-21 18:11:29,589 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ZooKeeper (Environment.java:logEnv(
98)) - Client environment:os.memory.max=871MB
2025-05-21 18:11:29,589 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ZooKeeper (Environment.java:logEnv(
98)) - Client environment:os.memory.total=106MB
2025-05-21 18:11:29,614 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ZooKeeper (ZooKeeper.java:<init>(63
7)) - Initiating client connection, connectString=127.0.0.1:2181 sessionTimeout=90000 watcher=org.apache.hadoop.hbase.zo
okeeper.ReadOnlyZKClient$$Lambda$22/970818114@11df9173
2025-05-21 18:11:29,628 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ClientCnxnSocket (ClientCnxnSocket.
java:initProperties(239)) - jute.maxbuffer value is 1048575 Bytes
2025-05-21 18:11:29,690 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6] zookeeper.ClientCnxn (ClientCnxn.java:initReq
uestTimeout(1747)) - zookeeper.request.timeout value is 0. feature enabled=false
2025-05-21 18:11:29,720 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:logStartConnect(1177)) - Opening socket connection to server localhost/127.0.0.1:2181.
2025-05-21 18:11:29,721 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:logStartConnect(1179)) - SASL config status: Will not attempt to authenticate using SASL (unknown er
ror)
2025-05-21 18:11:29,727 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:primeConnection(1013)) - Socket connection established, initiating session, client: /127.0.0.1:48548
, server: localhost/127.0.0.1:2181
2025-05-21 18:11:29,783 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x6892b3b6-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:onConnected(1453)) - Session establishment complete on server localhost/127.0.0.1:2181, session id =
 0x10000292f1b001f, negotiated timeout = 90000
Table Name: employees
Table Name: employees_copy
Table Name: employees_import
2025-05-21 18:11:32,768 INFO  [main] client.ConnectionImplementation (ConnectionImplementation.java:closeMasterService(2
127)) - Closing master protocol: MasterService
testtable is deleted
hduser@dist:~/data$
```

[ch 3-15]

```
2025-05-21 18:15:18,814 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x76a4ebf2] zookeeper.ZooKeeper (Environment.java:logEnv(
98)) - Client environment:user.dir=/home/hduser/data
2025-05-21 18:15:18,814 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x76a4ebf2] zookeeper.ZooKeeper (Environment.java:logEnv(
98)) - Client environment:os.memory.free=75MB
2025-05-21 18:15:18,815 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x76a4ebf2] zookeeper.ZooKeeper (Environment.java:logEnv(
98)) - Client environment:os.memory.max=871MB
2025-05-21 18:15:18,815 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x76a4ebf2] zookeeper.ZooKeeper (Environment.java:logEnv(
98)) - Client environment:os.memory.total=106MB
2025-05-21 18:15:18,820 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x76a4ebf2] zookeeper.ZooKeeper (ZooKeeper.java:<init>(63
7)) - Initiating client connection, connectString=127.0.0.1:2181 sessionTimeout=90000 watcher=org.apache.hadoop.hbase.zo
okeeper.ReadOnlyZKClient$$Lambda$22/1614869998@a2861b0
2025-05-21 18:15:18,830 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x76a4ebf2] zookeeper.ClientCnxnSocket (ClientCnxnSocket.
java:initProperties(239)) - jute.maxbuffer value is 1048575 Bytes
2025-05-21 18:15:18,843 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x76a4ebf2] zookeeper.ClientCnxn (ClientCnxn.java:initReq
uestTimeout(1747)) - zookeeper.request.timeout value is 0. feature enabled=false
2025-05-21 18:15:18,862 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x76a4ebf2-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:logStartConnect(1177)) - Opening socket connection to server localhost/127.0.0.1:2181.
2025-05-21 18:15:18,863 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x76a4ebf2-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:logStartConnect(1179)) - SASL config status: Will not attempt to authenticate using SASL (unknown er
ror)
2025-05-21 18:15:18,873 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x76a4ebf2-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:primeConnection(1013)) - Socket connection established, initiating session, client: /127.0.0.1:58898
, server: localhost/127.0.0.1:2181
2025-05-21 18:15:18,886 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x76a4ebf2-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:onConnected(1453)) - Session establishment complete on server localhost/127.0.0.1:2181, session id =
 0x10000292f1b0020, negotiated timeout = 90000
2025-05-21 18:15:21,464 INFO  [main] client.HBaseAdmin (HBaseAdmin.java:postOperationResult(3751)) - Operation: CREATE,
Table Name: default:random_table, procId: 32 completed
random_table : adding rows...
random_table is created
hduser@dist:~/data$
```

```
keyvalues={row-16/colfam1:col-1/1747851331716/Put/vlen=8/seqid=0}
keyvalues={row-17/colfam1:col-1/1747851332259/Put/vlen=8/seqid=0}
keyvalues={row-18/colfam1:col-1/1747851332784/Put/vlen=8/seqid=0}
keyvalues={row-19/colfam1:col-1/1747851333316/Put/vlen=8/seqid=0}
keyvalues={row-2/colfam1:col-1/1747851322672/Put/vlen=7/seqid=0}
keyvalues={row-20/colfam1:col-1/1747851333867/Put/vlen=8/seqid=0}
keyvalues={row-21/colfam1:col-1/1747851334413/Put/vlen=8/seqid=0}
keyvalues={row-22/colfam1:col-1/1747851335046/Put/vlen=8/seqid=0}
Scanning table #2... equal [.*-.5] regular expression
keyvalues={row-15/colfam1:col-1/1747851331336/Put/vlen=8/seqid=0}
keyvalues={row-25/colfam1:col-1/1747851336546/Put/vlen=8/seqid=0}
keyvalues={row-35/colfam1:col-1/1747851341308/Put/vlen=8/seqid=0}
keyvalues={row-45/colfam1:col-1/1747851345879/Put/vlen=8/seqid=0}
keyvalues={row-55/colfam1:col-1/1747851350358/Put/vlen=8/seqid=0}
keyvalues={row-65/colfam1:col-1/1747851354862/Put/vlen=8/seqid=0}
keyvalues={row-75/colfam1:col-1/1747851358866/Put/vlen=8/seqid=0}
keyvalues={row-85/colfam1:col-1/1747851362818/Put/vlen=8/seqid=0}
keyvalues={row-95/colfam1:col-1/1747851367065/Put/vlen=8/seqid=0}
Scanning table #3... equal substring {-5}
keyvalues={row-5/colfam1:col-1/1747851324998/Put/vlen=7/seqid=0}
keyvalues={row-50/colfam1:col-1/1747851348070/Put/vlen=8/seqid=0}
keyvalues={row-51/colfam1:col-1/1747851348567/Put/vlen=8/seqid=0}
keyvalues={row-52/colfam1:col-1/1747851349070/Put/vlen=8/seqid=0}
keyvalues={row-53/colfam1:col-1/1747851349514/Put/vlen=8/seqid=0}
keyvalues={row-54/colfam1:col-1/1747851349969/Put/vlen=8/seqid=0}
keyvalues={row-55/colfam1:col-1/1747851350358/Put/vlen=8/seqid=0}
keyvalues={row-56/colfam1:col-1/1747851350867/Put/vlen=8/seqid=0}
keyvalues={row-57/colfam1:col-1/1747851351209/Put/vlen=8/seqid=0}
keyvalues={row-58/colfam1:col-1/1747851351834/Put/vlen=8/seqid=0}
keyvalues={row-59/colfam1:col-1/1747851352254/Put/vlen=8/seqid=0}
hduser@dist:~/data$
```

```
keyvalues={row-76/colfam1:col-100/1747851359634/Put/vlen=10/seqid=0}
keyvalues={row-77/colfam1:col-100/1747851359967/Put/vlen=10/seqid=0}
keyvalues={row-78/colfam1:col-100/1747851360371/Put/vlen=10/seqid=0}
keyvalues={row-79/colfam1:col-100/1747851360729/Put/vlen=10/seqid=0}
keyvalues={row-8/colfam1:col-100/1747851327627/Put/vlen=9/seqid=0}
keyvalues={row-80/colfam1:col-100/1747851361136/Put/vlen=10/seqid=0}
keyvalues={row-81/colfam1:col-100/1747851361497/Put/vlen=10/seqid=0}
keyvalues={row-82/colfam1:col-100/1747851361884/Put/vlen=10/seqid=0}
keyvalues={row-83/colfam1:col-100/1747851362377/Put/vlen=10/seqid=0}
keyvalues={row-84/colfam1:col-100/1747851362815/Put/vlen=10/seqid=0}
keyvalues={row-85/colfam1:col-100/1747851363221/Put/vlen=10/seqid=0}
keyvalues={row-86/colfam1:col-100/1747851363633/Put/vlen=10/seqid=0}
keyvalues={row-87/colfam1:col-100/1747851364152/Put/vlen=10/seqid=0}
keyvalues={row-88/colfam1:col-100/1747851364556/Put/vlen=10/seqid=0}
keyvalues={row-89/colfam1:col-100/1747851365030/Put/vlen=10/seqid=0}
keyvalues={row-9/colfam1:col-100/1747851328134/Put/vlen=9/seqid=0}
keyvalues={row-90/colfam1:col-100/1747851365444/Put/vlen=10/seqid=0}
keyvalues={row-91/colfam1:col-100/1747851365824/Put/vlen=10/seqid=0}
keyvalues={row-92/colfam1:col-100/1747851366211/Put/vlen=10/seqid=0}
keyvalues={row-93/colfam1:col-100/1747851366623/Put/vlen=10/seqid=0}
keyvalues={row-94/colfam1:col-100/1747851367062/Put/vlen=10/seqid=0}
keyvalues={row-95/colfam1:col-100/1747851367464/Put/vlen=10/seqid=0}
keyvalues={row-96/colfam1:col-100/1747851367869/Put/vlen=10/seqid=0}
keyvalues={row-97/colfam1:col-100/1747851368288/Put/vlen=10/seqid=0}
keyvalues={row-98/colfam1:col-100/1747851368681/Put/vlen=10/seqid=0}
keyvalues={row-99/colfam1:col-100/1747851369140/Put/vlen=10/seqid=0}
##############################
Result of Get(): keyvalues={row-5/colfam1:col-100/1747851325708/Put/vlen=9/seqid=0}
##################################
Result of get(): keyvalues=NONE
hduser@dist:~/data$
```

```
m2:col-16/1747851368346/Put/vlen=9/seqid=0, row-98/colfam2:col-17/1747851368349/Put/vlen=9/seqid=0, row-98/colfam2:col-1
8/1747851368352/Put/vlen=9/seqid=0, row-98/colfam2:col-19/1747851368357/Put/vlen=9/seqid=0, row-98/colfam2:col-2/1747851
368296/Put/vlen=8/seqid=0}
keyvalues={row-99/colfam1:col-1/1747851368686/Put/vlen=8/seqid=0, row-99/colfam1:col-10/1747851368720/Put/vlen=9/seqid=0
, row-99/colfam1:col-100/1747851369140/Put/vlen=10/seqid=0, row-99/colfam1:col-11/1747851368727/Put/vlen=9/seqid=0, row-
99/colfam1:col-12/1747851368730/Put/vlen=9/seqid=0, row-99/colfam1:col-13/1747851368734/Put/vlen=9/seqid=0, row-99/colfa
m1:col-14/1747851368737/Put/vlen=9/seqid=0, row-99/colfam1:col-15/1747851368741/Put/vlen=9/seqid=0, row-99/colfam1:col-1
6/1747851368746/Put/vlen=9/seqid=0, row-99/colfam1:col-17/1747851368751/Put/vlen=9/seqid=0, row-99/colfam1:col-18/174785
1368758/Put/vlen=9/seqid=0, row-99/colfam1:col-19/1747851368763/Put/vlen=9/seqid=0, row-99/colfam1:col-2/1747851368689/P
ut/vlen=8/seqid=0, row-99/colfam2:col-1/1747851368686/Put/vlen=8/seqid=0, row-99/colfam2:col-10/1747851368720/Put/vlen=9
/seqid=0, row-99/colfam2:col-100/1747851369140/Put/vlen=10/seqid=0, row-99/colfam2:col-11/1747851368727/Put/vlen=9/seqid
=0, row-99/colfam2:col-12/1747851368730/Put/vlen=9/seqid=0, row-99/colfam2:col-13/1747851368734/Put/vlen=9/seqid=0, row-
99/colfam2:col-14/1747851368737/Put/vlen=9/seqid=0, row-99/colfam2:col-15/1747851368741/Put/vlen=9/seqid=0, row-99/colfa
m2:col-16/1747851368746/Put/vlen=9/seqid=0, row-99/colfam2:col-17/1747851368751/Put/vlen=9/seqid=0, row-99/colfam2:col-1
8/1747851368758/Put/vlen=9/seqid=0, row-99/colfam2:col-19/1747851368763/Put/vlen=9/seqid=0, row-99/colfam2:col-2/1747851
368689/Put/vlen=8/seqid=0}
##############################
Result of Get(): keyvalues={row-5/colfam1:col-1/1747851324998/Put/vlen=7/seqid=0, row-5/colfam1:col-10/1747851325053/Put
/vlen=8/seqid=0, row-5/colfam1:col-100/1747851325708/Put/vlen=9/seqid=0, row-5/colfam1:col-11/1747851325059/Put/vlen=8/s
eqid=0, row-5/colfam1:col-12/1747851325066/Put/vlen=8/seqid=0, row-5/colfam1:col-13/1747851325073/Put/vlen=8/seqid=0, ro
w-5/colfam1:col-14/1747851325077/Put/vlen=8/seqid=0, row-5/colfam1:col-15/1747851325085/Put/vlen=8/seqid=0, row-5/colfam
1:col-16/1747851325093/Put/vlen=8/seqid=0, row-5/colfam1:col-17/1747851325101/Put/vlen=8/seqid=0, row-5/colfam1:col-18/1
747851325108/Put/vlen=8/seqid=0, row-5/colfam1:col-19/1747851325114/Put/vlen=8/seqid=0, row-5/colfam1:col-2/174785132500
4/Put/vlen=7/seqid=0, row-5/colfam2:col-1/1747851324998/Put/vlen=7/seqid=0, row-5/colfam2:col-10/1747851325053/Put/vlen=
8/seqid=0, row-5/colfam2:col-100/1747851325708/Put/vlen=9/seqid=0, row-5/colfam2:col-11/1747851325059/Put/vlen=8/seqid=0
, row-5/colfam2:col-12/1747851325066/Put/vlen=8/seqid=0, row-5/colfam2:col-13/1747851325073/Put/vlen=8/seqid=0, row-5/co
lfam2:col-14/1747851325077/Put/vlen=8/seqid=0, row-5/colfam2:col-15/1747851325085/Put/vlen=8/seqid=0, row-5/colfam2:col-
16/1747851325093/Put/vlen=8/seqid=0, row-5/colfam2:col-17/1747851325101/Put/vlen=8/seqid=0, row-5/colfam2:col-18/1747851
325108/Put/vlen=8/seqid=0, row-5/colfam2:col-19/1747851325114/Put/vlen=8/seqid=0, row-5/colfam2:col-2/1747851325004/Put/
vlen=7/seqid=0}
hduser@dist:~/data$
```

[ch 3-16]

```
2025-05-21 18:18:19,859 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x7d0b7e3c] zookeeper.ClientCnxnSocket (ClientCnxnSocket.
java:initProperties(239)) - jute.maxbuffer value is 1048575 Bytes
2025-05-21 18:18:19,875 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x7d0b7e3c] zookeeper.ClientCnxn (ClientCnxn.java:initReq
uestTimeout(1747)) - zookeeper.request.timeout value is 0. feature enabled=false
2025-05-21 18:18:19,895 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x7d0b7e3c-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:logStartConnect(1177)) - Opening socket connection to server localhost/127.0.0.1:2181.
2025-05-21 18:18:19,895 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x7d0b7e3c-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:logStartConnect(1179)) - SASL config status: Will not attempt to authenticate using SASL (unknown er
ror)
2025-05-21 18:18:19,905 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x7d0b7e3c-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:primeConnection(1013)) - Socket connection established, initiating session, client: /127.0.0.1:37692
, server: localhost/127.0.0.1:2181
2025-05-21 18:18:19,916 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x7d0b7e3c-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:onConnected(1453)) - Session establishment complete on server localhost/127.0.0.1:2181, session id =
 0x10000292f1b0025, negotiated timeout = 90000
Scanning table #1...equal colfam1, col-100, val-5
Cell: row-5/colfam1:col-100/1747851325708/Put/vlen=9/seqid=0, Value: val-5.100
Cell: row-50/colfam1:col-100/1747851348558/Put/vlen=10/seqid=0, Value: val-50.100
Cell: row-51/colfam1:col-100/1747851349067/Put/vlen=10/seqid=0, Value: val-51.100
Cell: row-52/colfam1:col-100/1747851349508/Put/vlen=10/seqid=0, Value: val-52.100
Cell: row-53/colfam1:col-100/1747851349965/Put/vlen=10/seqid=0, Value: val-53.100
Cell: row-54/colfam1:col-100/1747851350355/Put/vlen=10/seqid=0, Value: val-54.100
Cell: row-55/colfam1:col-100/1747851350863/Put/vlen=10/seqid=0, Value: val-55.100
Cell: row-56/colfam1:col-100/1747851351205/Put/vlen=10/seqid=0, Value: val-56.100
Cell: row-57/colfam1:col-100/1747851351830/Put/vlen=10/seqid=0, Value: val-57.100
Cell: row-58/colfam1:col-100/1747851352250/Put/vlen=10/seqid=0, Value: val-58.100
Cell: row-59/colfam1:col-100/1747851352690/Put/vlen=10/seqid=0, Value: val-59.100
################################
Result of Get(): row-59
Cell: row-59/colfam1:col-100/1747851352690/Put/vlen=10/seqid=0, Value: val-59.100
hduser@dist:~/data$
```

```
Cell: row-100/colfam2:col-73/1747851369459/Put/vlen=10/seqid=0, Value: val-100.73
Cell: row-100/colfam2:col-74/1747851369463/Put/vlen=10/seqid=0, Value: val-100.74
Cell: row-100/colfam2:col-75/1747851369466/Put/vlen=10/seqid=0, Value: val-100.75
Cell: row-100/colfam2:col-76/1747851369470/Put/vlen=10/seqid=0, Value: val-100.76
Cell: row-100/colfam2:col-77/1747851369474/Put/vlen=10/seqid=0, Value: val-100.77
Cell: row-100/colfam2:col-78/1747851369478/Put/vlen=10/seqid=0, Value: val-100.78
Cell: row-100/colfam2:col-79/1747851369482/Put/vlen=10/seqid=0, Value: val-100.79
Cell: row-100/colfam2:col-8/1747851369178/Put/vlen=9/seqid=0, Value: val-100.8
Cell: row-100/colfam2:col-80/1747851369486/Put/vlen=10/seqid=0, Value: val-100.80
Cell: row-100/colfam2:col-81/1747851369489/Put/vlen=10/seqid=0, Value: val-100.81
Cell: row-100/colfam2:col-82/1747851369497/Put/vlen=10/seqid=0, Value: val-100.82
Cell: row-100/colfam2:col-83/1747851369505/Put/vlen=10/seqid=0, Value: val-100.83
Cell: row-100/colfam2:col-84/1747851369510/Put/vlen=10/seqid=0, Value: val-100.84
Cell: row-100/colfam2:col-85/1747851369514/Put/vlen=10/seqid=0, Value: val-100.85
Cell: row-100/colfam2:col-86/1747851369518/Put/vlen=10/seqid=0, Value: val-100.86
Cell: row-100/colfam2:col-87/1747851369522/Put/vlen=10/seqid=0, Value: val-100.87
Cell: row-100/colfam2:col-88/1747851369525/Put/vlen=10/seqid=0, Value: val-100.88
Cell: row-100/colfam2:col-89/1747851369528/Put/vlen=10/seqid=0, Value: val-100.89
Cell: row-100/colfam2:col-9/1747851369183/Put/vlen=9/seqid=0, Value: val-100.9
Cell: row-100/colfam2:col-90/1747851369532/Put/vlen=10/seqid=0, Value: val-100.90
Cell: row-100/colfam2:col-91/1747851369539/Put/vlen=10/seqid=0, Value: val-100.91
Cell: row-100/colfam2:col-92/1747851369545/Put/vlen=10/seqid=0, Value: val-100.92
Cell: row-100/colfam2:col-93/1747851369550/Put/vlen=10/seqid=0, Value: val-100.93
Cell: row-100/colfam2:col-94/1747851369553/Put/vlen=10/seqid=0, Value: val-100.94
Cell: row-100/colfam2:col-95/1747851369556/Put/vlen=10/seqid=0, Value: val-100.95
Cell: row-100/colfam2:col-96/1747851369560/Put/vlen=10/seqid=0, Value: val-100.96
Cell: row-100/colfam2:col-97/1747851369564/Put/vlen=10/seqid=0, Value: val-100.97
Cell: row-100/colfam2:col-98/1747851369569/Put/vlen=10/seqid=0, Value: val-100.98
Cell: row-100/colfam2:col-99/1747851369573/Put/vlen=10/seqid=0, Value: val-100.99
################################
Result of Get(): row-59
```

```
14: keyvalues={row-76/colfam1:col-100/1747851359634/Put/vlen=10/seqid=0}
start row: row-76\x00
0: keyvalues={row-77/colfam1:col-100/1747851359967/Put/vlen=10/seqid=0}
1: keyvalues={row-78/colfam1:col-100/1747851360371/Put/vlen=10/seqid=0}
2: keyvalues={row-79/colfam1:col-100/1747851360729/Put/vlen=10/seqid=0}
3: keyvalues={row-8/colfam1:col-100/1747851327627/Put/vlen=9/seqid=0}
4: keyvalues={row-80/colfam1:col-100/1747851361136/Put/vlen=10/seqid=0}
5: keyvalues={row-81/colfam1:col-100/1747851361497/Put/vlen=10/seqid=0}
6: keyvalues={row-82/colfam1:col-100/1747851361884/Put/vlen=10/seqid=0}
7: keyvalues={row-83/colfam1:col-100/1747851362377/Put/vlen=10/seqid=0}
8: keyvalues={row-84/colfam1:col-100/1747851362815/Put/vlen=10/seqid=0}
9: keyvalues={row-85/colfam1:col-100/1747851363221/Put/vlen=10/seqid=0}
10: keyvalues={row-86/colfam1:col-100/1747851363633/Put/vlen=10/seqid=0}
11: keyvalues={row-87/colfam1:col-100/1747851364152/Put/vlen=10/seqid=0}
12: keyvalues={row-88/colfam1:col-100/1747851364556/Put/vlen=10/seqid=0}
13: keyvalues={row-89/colfam1:col-100/1747851365030/Put/vlen=10/seqid=0}
14: keyvalues={row-9/colfam1:col-100/1747851328134/Put/vlen=9/seqid=0}
start row: row-9\x00
0: keyvalues={row-90/colfam1:col-100/1747851365444/Put/vlen=10/seqid=0}
1: keyvalues={row-91/colfam1:col-100/1747851365824/Put/vlen=10/seqid=0}
2: keyvalues={row-92/colfam1:col-100/1747851366211/Put/vlen=10/seqid=0}
3: keyvalues={row-93/colfam1:col-100/1747851366623/Put/vlen=10/seqid=0}
4: keyvalues={row-94/colfam1:col-100/1747851367062/Put/vlen=10/seqid=0}
5: keyvalues={row-95/colfam1:col-100/1747851367464/Put/vlen=10/seqid=0}
6: keyvalues={row-96/colfam1:col-100/1747851367869/Put/vlen=10/seqid=0}
7: keyvalues={row-97/colfam1:col-100/1747851368288/Put/vlen=10/seqid=0}
8: keyvalues={row-98/colfam1:col-100/1747851368681/Put/vlen=10/seqid=0}
9: keyvalues={row-99/colfam1:col-100/1747851369140/Put/vlen=10/seqid=0}
start row: row-99\x00
total rows: 100
hduser@dist:~/data$
```

```
2025-05-21 18:19:25,149 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x1530c739-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:primeConnection(1013)) - Socket connection established, initiating session, client: /127.0.0.1:42586
, server: localhost/127.0.0.1:2181
2025-05-21 18:19:25,180 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x1530c739-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:onConnected(1453)) - Session establishment complete on server localhost/127.0.0.1:2181, session id =
 0x10000292f1b0028, negotiated timeout = 90000
Scanning table #1... row-3 ~ row-5
Cell: row-3/colfam1:col-100/1747851324261/Put/vlen=9/seqid=0, Value: val-3.100
Cell: row-30/colfam1:col-100/1747851339459/Put/vlen=10/seqid=0, Value: val-30.100
Cell: row-31/colfam1:col-100/1747851339946/Put/vlen=10/seqid=0, Value: val-31.100
Cell: row-32/colfam1:col-100/1747851340450/Put/vlen=10/seqid=0, Value: val-32.100
Cell: row-33/colfam1:col-100/1747851340995/Put/vlen=10/seqid=0, Value: val-33.100
Cell: row-34/colfam1:col-100/1747851341305/Put/vlen=10/seqid=0, Value: val-34.100
Cell: row-35/colfam1:col-100/1747851341670/Put/vlen=10/seqid=0, Value: val-35.100
Cell: row-36/colfam1:col-100/1747851342121/Put/vlen=10/seqid=0, Value: val-36.100
Cell: row-37/colfam1:col-100/1747851342576/Put/vlen=10/seqid=0, Value: val-37.100
Cell: row-38/colfam1:col-100/1747851343038/Put/vlen=10/seqid=0, Value: val-38.100
Cell: row-39/colfam1:col-100/1747851343532/Put/vlen=10/seqid=0, Value: val-39.100
Cell: row-4/colfam1:col-100/1747851324991/Put/vlen=9/seqid=0, Value: val-4.100
Cell: row-40/colfam1:col-100/1747851343943/Put/vlen=10/seqid=0, Value: val-40.100
Cell: row-41/colfam1:col-100/1747851344397/Put/vlen=10/seqid=0, Value: val-41.100
Cell: row-42/colfam1:col-100/1747851344890/Put/vlen=10/seqid=0, Value: val-42.100
Cell: row-43/colfam1:col-100/1747851345410/Put/vlen=10/seqid=0, Value: val-43.100
Cell: row-44/colfam1:col-100/1747851345874/Put/vlen=10/seqid=0, Value: val-44.100
Cell: row-45/colfam1:col-100/1747851346351/Put/vlen=10/seqid=0, Value: val-45.100
Cell: row-46/colfam1:col-100/1747851346775/Put/vlen=10/seqid=0, Value: val-46.100
Cell: row-47/colfam1:col-100/1747851347200/Put/vlen=10/seqid=0, Value: val-47.100
Cell: row-48/colfam1:col-100/1747851347614/Put/vlen=10/seqid=0, Value: val-48.100
Cell: row-49/colfam1:col-100/1747851348066/Put/vlen=10/seqid=0, Value: val-49.100
Cell: row-5/colfam1:col-100/1747851325708/Put/vlen=9/seqid=0, Value: val-5.100
hduser@dist:~/data$
```

```
2025-05-21 18:19:37,221 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x78e4deb0] zookeeper.ZooKeeper (Environment.java:logEnv(
98)) - Client environment:user.home=/home/hduser
2025-05-21 18:19:37,221 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x78e4deb0] zookeeper.ZooKeeper (Environment.java:logEnv(
98)) - Client environment:user.dir=/home/hduser/data
2025-05-21 18:19:37,222 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x78e4deb0] zookeeper.ZooKeeper (Environment.java:logEnv(
98)) - Client environment:os.memory.free=74MB
2025-05-21 18:19:37,222 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x78e4deb0] zookeeper.ZooKeeper (Environment.java:logEnv(
98)) - Client environment:os.memory.max=871MB
2025-05-21 18:19:37,223 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x78e4deb0] zookeeper.ZooKeeper (Environment.java:logEnv(
98)) - Client environment:os.memory.total=106MB
2025-05-21 18:19:37,231 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x78e4deb0] zookeeper.ZooKeeper (ZooKeeper.java:<init>(63
7)) - Initiating client connection, connectString=127.0.0.1:2181 sessionTimeout=90000 watcher=org.apache.hadoop.hbase.zo
okeeper.ReadOnlyZKClient$$Lambda$22/902758381@2f0fbb5c
2025-05-21 18:19:37,238 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x78e4deb0] zookeeper.ClientCnxnSocket (ClientCnxnSocket.
java:initProperties(239)) - jute.maxbuffer value is 1048575 Bytes
2025-05-21 18:19:37,252 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x78e4deb0] zookeeper.ClientCnxn (ClientCnxn.java:initReq
uestTimeout(1747)) - zookeeper.request.timeout value is 0. feature enabled=false
2025-05-21 18:19:37,273 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x78e4deb0-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:logStartConnect(1177)) - Opening socket connection to server localhost/127.0.0.1:2181.
2025-05-21 18:19:37,273 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x78e4deb0-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:logStartConnect(1179)) - SASL config status: Will not attempt to authenticate using SASL (unknown er
ror)
2025-05-21 18:19:37,281 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x78e4deb0-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:primeConnection(1013)) - Socket connection established, initiating session, client: /127.0.0.1:42308
, server: localhost/127.0.0.1:2181
2025-05-21 18:19:37,310 INFO  [ReadOnlyZKClient-127.0.0.1:2181@0x78e4deb0-SendThread(127.0.0.1:2181)] zookeeper.ClientCn
xn (ClientCnxn.java:onConnected(1453)) - Session establishment complete on server localhost/127.0.0.1:2181, session id =
 0x10000292f1b0029, negotiated timeout = 90000
Scanning table #1... ts match
Results of scan #2: ts range
hduser@dist:~/data$
```